Título do trabalho

um subtítulo

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Report presented to the
Institute of Mathematics,
Statistics, and Computer Science
of the University of São Paulo
for the Doctor of Science
qualifying examination

Program: Computer Science
Advisor: Prof. Dr. Alfredo Goldman

During the development of this work, the author received financial support from FAPESP under grant #2023/09048-8. This study was also financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) – Finance Code 001.

São Paulo
September, 2026

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This is the original version of the
qualifying text prepared by candidate
Lucas de Sousa Rosa, as submitted
to the Examining Committee.

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Chapter 1

Introduction

High-performance computing (HPC) has become essential to modern science, and its capability has grown by drawing more power, packing more hardware, and cycling through that hardware ever faster (Portegies Zwart, 2020; Allen, 2022). The environmental cost of this growth has traditionally been equated with operational energy consumption (Koomey *et al.*, 2011), yet operational energy is only one facet of the footprint. Manufacturing the processors, memory, and interconnects that populate a machine emits carbon long before the system is switched on, and this embodied share is rising as fabrication grows more complex (Gupta *et al.*, 2021; Pirson *et al.*, 2023). Cooling those systems and generating the electricity that feeds them consumes freshwater at scales large enough to matter regionally (Mytton, 2021). The semiconductor supply chain draws on scarce materials and discharges hazardous waste (Sandhu *et al.*, 2025), while the shortening replacement cycles that keep systems competitive turn still-functional hardware into electronic waste. Across its full lifecycle, then, an HPC system carries costs in carbon, water, and materials and leaves electronic waste behind, well beyond the operational energy it draws.

Visibility into this broader footprint has only recently begun to grow, and the established ways of evaluating and operating HPC systems still capture only part of it. The Green500 ranks systems by performance per watt under a single benchmark (Feng and Cameron, 2007), reducing sustainability to an efficiency ratio measured on Linpack. Job schedulers decide when and where computation runs, yet they have conventionally been designed to prioritize performance above all else (Kocot *et al.*, 2023; Czarnul *et al.*, 2019), leaving environmental cost outside the optimization objective. Procurement and hardware-replacement decisions weigh performance and acquisition cost while treating lifecycle impact as an afterthought (Li *et al.*, 2023). Each of these approaches sharpens a single dimension of a problem whose full extent remains only partially charted, and the gains they deliver are repeatedly absorbed by growing demand, so the best-measured dimension does not improve in absolute terms (York *et al.*, 2022). Attention to the dimensions these approaches leave out is a comparatively recent and still-emerging area of interest.

This work begins with a systematic mapping study that turns this emerging interest into a concrete gap (Rosa *et al.*, 2026). Classifying 62 reports from sixteen years of research across environmental impacts, lifecycle stages, system loci, and management levers, the study finds the literature concentrated in a single corridor of that classification: carbon among the impacts, operations among the lifecycle stages, facility infrastructure among the system loci, and measurement among the management levers. The dimensions outside that corridor sit in sparsely populated cells, studied rarely and in isolation: water beyond cooling, materials beyond what carbon accounting already captures, biodiversity, and end-of-life processing. The rest of this work sets out to fill the gaps the map identifies, and it currently runs along two branches.

The first branch focuses on HPC systems' operation. The batch scheduler decides when and where jobs run on performance grounds, users submit work without seeing its environmental cost, and hardware-replacement decisions weigh performance and price while overlooking embodied and end-of-life impacts. This work responds by building simulation infrastructure to test a range of mechanisms: scheduling heuristics that react to a time-varying grid, user-facing accounting that makes cost visible, platform controls that trade performance for a smaller footprint, and lifecycle-aware policies that tie hardware aging and replacement to an environmental budget. The infrastructure and its footprint model track operational carbon, water, and embodied impact, so this work can study these mechanisms without access to a production system. A first result along this line is Greenfilling, a small change to EASY backfilling (Skovira *et al.*, 1996) that holds back opportunistic job starts when the targeted grid intensity, carbon or water, runs above its recent average. Evaluated across three national grids and two production machines, it bought certain delay for uncertain environmental return, which sets a lower bound on how sophisticated such an intervention has to be. A job shifting variant was also developed. It prices delaying each job against the grid intensities it would meet and an offline analysis of the same workloads and signals places its ceiling near 7% of operational carbon, two orders of magnitude above what Greenfilling reached.

The second branch focuses on HPC systems' impacts as a whole. The Green500 ranks systems by performance per watt, but this is just one side of the real impacts of using those platforms. Two systems can rank side by side while drawing very different absolute power, sitting on grids whose water and carbon intensity differ by orders of magnitude, carrying different embodied impacts, and consuming water at rates that do not match the regional scarcity. The aim is to use Green500 as a reference point, showing what a sufficiency-aware lens reveals: cases where efficiency improves while absolute burden grows, where embodied costs dominate but stay invisible, and where regional context changes which system is actually less harmful.

The remainder of this document is organized as follows. Chapter 2 reviews the environmental footprint of HPC across its lifecycle, then presents the systematic mapping study and the classification behind the gap this work addresses. Chapter 3 reports the preliminary results. Chapter 4 sets out what remains. Add user modeling, platform controls, and lifecycle-aware policies to the testbed built in the first branch. Design a sufficiency-aware evaluation that uses the Green500 as a reference point, widening the ranking to the impacts efficiency omits and reaching toward the absolute burden a per-watt ratio hides, as far as public data and what-if scenarios permit. Chapter 5 closes with the schedule and the expected contributions of the completed work.

Chapter 2

The Environmental Impacts of HPC

Before tackling the problem, one needs to understand how existing literature sees it. A systematic mapping study (SMS) enables a quick scan across a body of work to answer questions such as what has been studied, how it has been studied, and where the literature runs thin. This chapter presents what the map of the environmental impacts of high-performance computing looks like. The SMS was conducted following the guidelines of Petersen et al. (Petersen *et al.*, 2008).

2.1 What the Existing Reviews Miss

What is the current landscape, how is it addressed, and where are the blind spots? These are the questions we sought answers to. However, existing reviews do not answer them in a satisfactory manner.

Most of the existing work looks at operational energy. Reviews of HPC power and energy stay close to the running machine. O'Brien et al. (O'brien *et al.*, 2018) survey predictive power and energy models for heterogeneous architectures, Czarnul et al. (Czarnul *et al.*, 2019) catalog energy-aware tools such as DVFS and power capping, Kocot et al. (Kocot *et al.*, 2023) review energy-aware scheduling, and Cai et al. (Cai *et al.*, 2011) organize the field around the trade-off between power and performance. Silva et al. (Silva *et al.*, 2024) reach further, integrating carbon and renewable management across infrastructure, hardware, software, and policy, yet the analysis still turns on the operational energy-carbon nexus and leaves the wider environmental picture to future work. These reviews map operational energy well. They do not set out to cover several environmental impacts, or to follow a system across its lifecycle.

The broader sustainability literature on data centers repeats the pattern. Surveys there cover renewable integration and waste heat (Shuja *et al.*, 2016), cooling systems (Zhang *et al.*, 2021), thermal-aware scheduling (Chaudhry *et al.*, 2015), and operational metrics such as PUE and WUE (Nagarathinam *et al.*, 2024), and the most recent ones add explicit review protocols and richer operational measures (Kahil *et al.*, 2025; Safari *et al.*, 2025; Kumar and Pindoriya, 2026). The scope stays on operational efficiency, and the setting is the data center rather than HPC.

A few reviews break out of that frame. Whitehead et al. (Whitehead *et al.*, 2014; Whitehead *et al.*, 2015) show what a wider data-center assessment can hold, spanning energy, carbon, water, materials, human health, and waste disposal across the lifecycle, and D'Orgeval et al. (d'ORGEVAL *et al.*, 2024) propose a holistic framework that joins environmental, economic, and ecosystem concerns. These are the works that show how embodied and operational impacts interact, and why operational energy alone is a thin basis for environmental judgment. They also step outside HPC to do it.

Across the reviews we found, none brings all four together: HPC scope, a systematic protocol, several environmental impacts, and lifecycle coverage (Table 2.1). The broad, lifecycle-aware reviews tend to look past HPC, and the HPC-focused ones tend to stay close to operational energy, so relatively little work sits at their intersection. That gap is what motivates a systematic mapping study. It applies an explicit protocol to a question the existing reviews mostly leave open: how the environmental impacts of HPC are discussed across the lifecycle.

Reference	**Year**	**D1**	**D2**	**D3**	**D4**	**D5**	**D6**	**D7**	**D8**
Cai *et al.*	2011		✓					✓	
Whitehead *et al.*	2014			✓	✓	✓	✓	✓	✓
Whitehead *et al.*	2015			✓	✓	✓	✓	✓	✓
Chaudhry *et al.*	2015			✓				✓	
Shuja *et al.*	2016			✓				✓	
O'Brien *et al.*	2018		✓					✓	
Czarnul *et al.*	2019		✓					✓	
Zhang *et al.*	2021							✓	
Kocot *et al.*	2023		✓					✓	
Silva *et al.*	2024		✓	✓	✓	✓	✓	✓	✓
d'ORGEVAL *et al.*	2024			✓	✓	✓	✓	✓	✓
Nagarathinam *et al.*	2024			✓	✓	✓	✓	✓	
Kahil *et al.*	2025	✓		✓	✓			✓	
Safari *et al.*	2025	✓		✓	✓			✓	
Kumar and Pindoriya	2026							✓	
Our study	2026	✓	✓	✓	✓	✓	✓	✓	✓

D1 systematic protocol, D2 HPC scope, D3 carbon emissions, D4 water use, D5 materials and waste, D6 production stage, D7 operational stage, D8 end of life.

Table 2.1: *Related reviews and surveys compared along four dimensions: systematic methodology (D1), HPC scope (D2), breadth of environmental impacts (D3–D5), and lifecycle coverage (D6–D8). A review counts under D1 only if it documents a protocol with explicit inclusion and exclusion criteria and a screening process, following established guidelines such as those of Kitchenham et al. (Kitchenham and Charters, 2007) and Petersen et al. (Petersen et al., 2008).*

2.2 How the Map Was Drawn

The corpus was drawn from three bibliographic databases: IEEE Xplore, ACM Digital Library, and Scopus. The search query followed a Population, Intervention, Comparison, and Outcome rationale (Table 2.2), pairing terms for HPC systems and infrastructure with terms for environmental assessment and sustainability, and it ran over titles, abstracts, and keywords. Inclusion and exclusion criteria were fixed before any record was read (Table 2.3). The three databases returned 4812 records, and 3788 remained once duplicates were removed.

Element	Definition	Terms used
Population	HPC systems and infrastructure	*high-performance computing, supercomputing, supercomputer, hpc*
Intervention	Environmental assessment or sustainability practices	*sustainability, sustainable, ecological, footprint, environmental impact, carbon emission, greenhouse gas, water consumption, water usage, lifecycle assessment, lca, embodied carbon, e-waste, electronic waste, material depletion, resource consumption, rare earth*

Table 2.2: *PICO elements used to structure the SMS search string. Comparison and Outcome were not included, as they do not fit the purpose of the mapping.*

Code	Criterion
IC1	Reports addressing at least one environmental impact in the HPC context
IC2	Reports presenting methodologies for predicting or measuring the environmental impacts of HPC systems
EC1	Reports not related to the study scope
EC2	Reports focusing solely on energy without connecting to broader environmental impacts
EC3	Reports not in English, unavailable, or inaccessible

Table 2.3: *Selection criteria employed in this study.*

Screening these records paired a manually reviewed baseline with an automated pass. We first screened a subset by hand and extended it through backward and forward snowballing. Three large language models then voted independently on each record of the full corpus, and the records they agreed to include were decided by hand. The categories of the classification were not fixed in advance either. They were built inductively by keywording the abstracts of the included papers and grouping the terms that recurred. The full protocol, the database exports, the prompts, and the coding files are released with the study (Rosa *et al.*, 2026). The search closed on 11 April 2026 and left a final corpus of 62 reports covering sixteen years of research.

2.3 A Field Published Elsewhere

One property of that corpus is worth stating before the classification, because it shapes how the field sees itself. More than seventy percent of the reports appeared outside the core venues of the HPC community, in general computer science, in parallel and distributed systems, and in domain-specific journals.

The corpus is also young and uneven in time (Figure 2.1). Publication was sparse and intermittent from 2010 to 2021, only seventeen reports across twelve years, with several years contributing none at all. It then expanded sharply, with four reports in 2023, twelve in 2024, and twenty-four in 2025. The year 2026 is left out, since the search reached only its first months.

Where that growth landed is the telling part. Between 2023 and 2025, venues outside HPC took the largest share of it, ahead of both the HPC venues and the parallel and distributed ones. Much of the recent work therefore circulates where HPC researchers do not routinely look, which suggests the distance between where this research is produced and where it would consolidate is widening rather than closing.

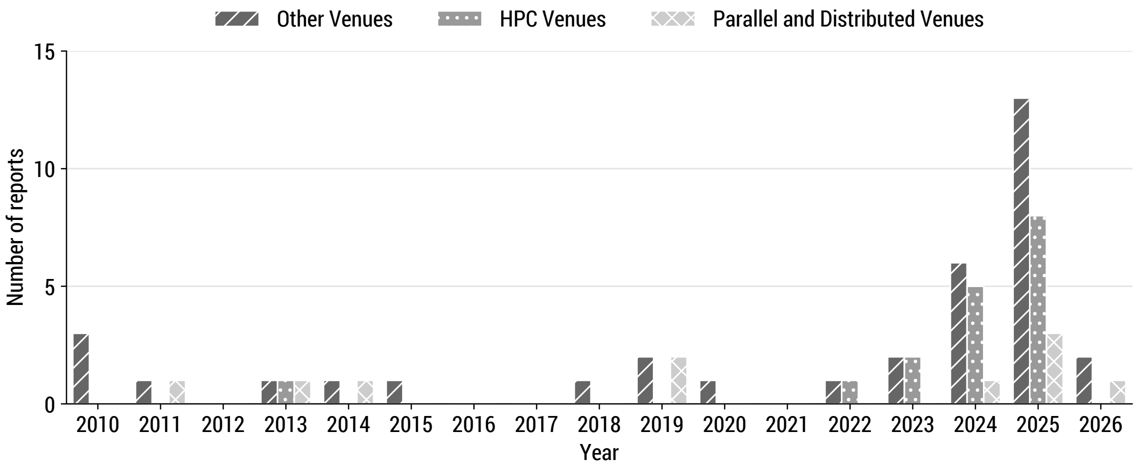

Figure 2.1: *Annual report counts in the final corpus, segmented by venue category.*

2.4 A Four-Facet Map of the Field

The keywording settled on four facets, and they carry the whole analysis (Figure 2.2). Each captures a different aspect of a paper. Environmental impacts identify the burden studied and divide into carbon emissions, materials and waste, water use, and biodiversity and ecosystems. Lifecycle stages locate that burden in the system's life, from production upstream, through operational use, to end of life. System loci place the work within the stack, from facility and energy infrastructure, through compute hardware, to software and workloads. Management and intervention levers describe how the impact is handled, from measurement and accounting toward workload and runtime optimization, infrastructure design, and incentives and policy. Three of the four facets are ordered rather than nominal, so a paper's place within them reads as a position on a scale, upstream to downstream, hardware to software, or measuring a problem to acting on it.

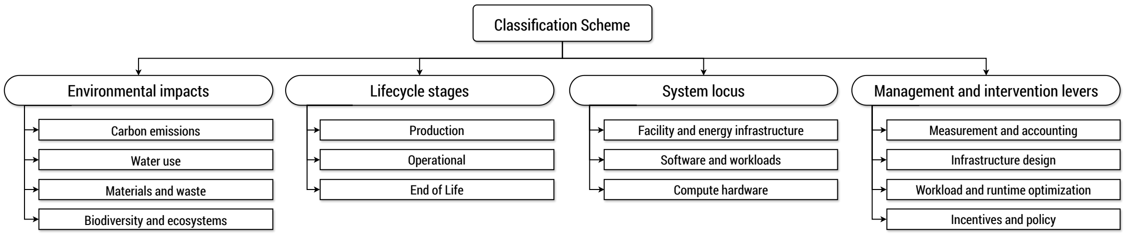

Figure 2.2: *The four-facet classification scheme derived from keywording the corpus. Each facet is subdivided into categories: environmental impacts, lifecycle stages, system loci, and management and intervention levers.*

Environmental impacts anchor the other three. Reading each of the remaining facets against it turns the classification into the literature map: which impact is studied at which stage, at which locus, and under which lever. Seen this way, the informative feature of the corpus is not that it has a dense region but how narrow that region is, and how much of the map stays empty.

2.5 The Corridor

The four impact categories are not studied in anything close to equal measure (Figure 2.3). Carbon emissions appear in 54 of the 62 reports. Materials and waste appear in 12, water use in 5, and biodiversity and ecosystems in a single report. The corpus also treats these burdens one at a time. Fifty-two reports address exactly one impact category, ten address two, and none reach three. The pairs that do occur are not arbitrary: nine of the ten two-impact reports combine carbon with materials, which share the same manufacturing inventories, and the one remaining pair joins carbon with water through cooling.

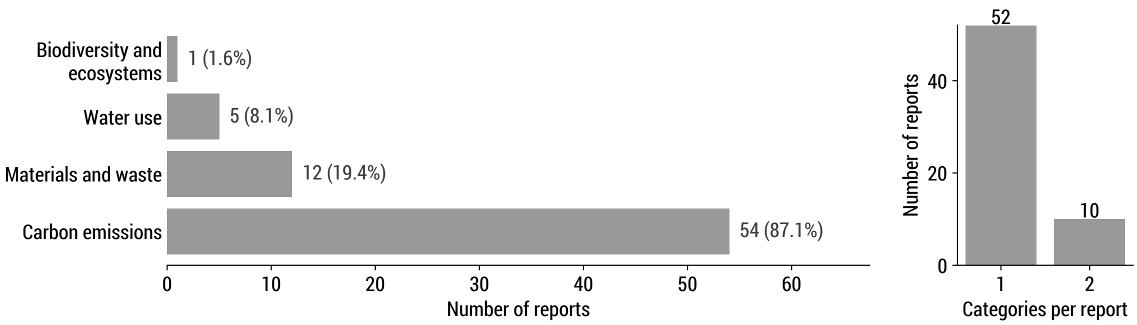

Figure 2.3: *Distribution of the 62 reports across the four environmental impact categories (left) and the number of categories each report addresses (right).*

The same concentration holds along the other three facets, and it lines up into a single corridor (Figure 2.4). The operational stage is the most populated point of the lifecycle for every impact category. Within carbon, the facility and energy infrastructure locus carries the most reports, 38 of them, ahead of software and workloads and of compute hardware.

Among the levers, measurement and accounting leads. The densest cells therefore share four coordinates, carbon studied at the operational stage, at the facility level, and through measurement, and the bulk of the literature clusters where those coordinates meet.

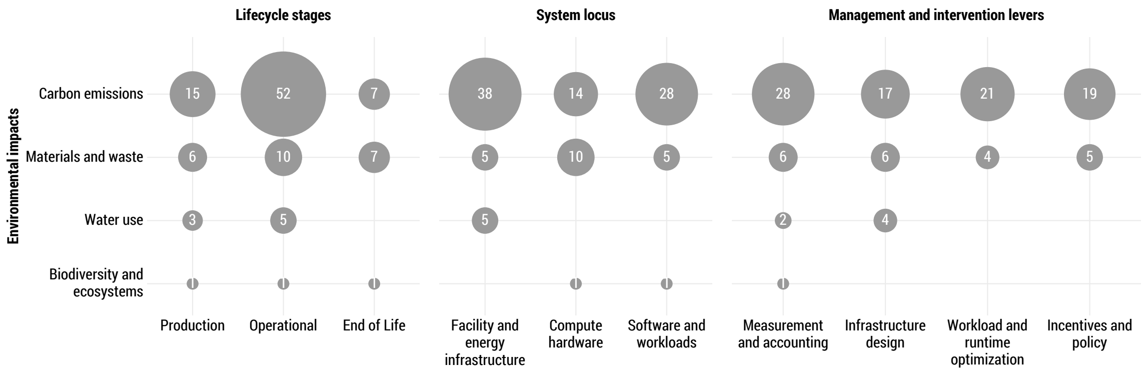

Figure 2.4: *Co-occurrence of the environmental impact categories (rows) with the lifecycle stages, system loci, and management and intervention levers (columns), across the 62 reports. Bubble size encodes the number of reports in each cell, and a missing bubble marks a combination no report addresses.*

Why the corridor falls where it does has less to do with which impacts matter and more with which are easy to instrument. Carbon is the readiest to quantify, because power telemetry and grid emission factors are widely available and external reporting gives an added reason to publish the numbers. Hölbling et al. (Hölbling *et al.*, 2025) use exactly these inputs to measure operational and embodied carbon across five Austrian universities and find a hundredfold variation in emissions per CPU hour with the regional electricity mix, and Rao et al. (Rao and Chien, 2025) model the carbon footprint of hundreds of Top500 systems to produce the first fleet-wide estimates. The operational stage produces continuous telemetry and job logs without extra instrumentation, the facility level offers meters and emission factors as a ready-made signal, and measurement is the lever the others build on. The corridor, in other words, traces the availability of data more than the shape of the problem.

2.6 The Periphery

Outside the corridor each minor category thins in its own way, and the empty cells are as telling as the full ones. Water use grew out of cooling engineering and has stayed there. Its five reports sit entirely at the facility level, from groundwater cooling models (Sheldon *et al.*, 2015) to thermosyphon designs that cut on-site consumption (Sickinger *et al.*, 2018), and only recently has the category begun to treat water as a footprint to model in its own right rather than a byproduct of cooling, accounting for the indirect water spent generating electricity and weighting it by regional scarcity (Jiang *et al.*, 2025). Water reaches neither compute hardware nor software, and it never appears at end of life, even though chip fabrication consumes large volumes of ultrapure water (Sandhu *et al.*, 2025) and disposal-stage leaching can contaminate groundwater (Houessionon *et al.*, 2021). Those burdens are real and documented, but not inside any HPC-specific study.

Materials and waste is the larger of the minor categories, yet it rarely stands on its own. Nine of its twelve reports also account for carbon, and the work concentrates at compute hardware because both impacts draw on the same manufacturing-stage inventories, so materials tend to ride on data already gathered for carbon accounting. Only a few papers make materials the primary object, among them a proposal for a metal usage effectiveness metric defined per unit of computation (Reinbigler and Martin, 2025). Materials do reach the end of life stage more often than the other minor categories, because electronic waste creates a tangible disposal pathway. Estimates put HPC e-waste disposal costs in the hundreds of millions of dollars within the decade (Zhao *et al.*, 2025), and several reports try to defer that cost by extending hardware life, whether by repurposing decommissioned machines (Lynar *et al.*, 2010) or by matching replacement cycles to the balance between embodied and operational carbon (Uwizeyimana and Jerger, 2025).

Biodiversity and ecosystems is the extreme case, present in a single report, a fab-to-grave analysis that measures embodied and operational biodiversity impact in species-year units and finds the operational share, driven by grid electricity, far larger than the embodied one (Shi *et al.*, 2025). The methods to produce a second such study exist in the broader environmental literature, including characterization factors that translate resource use into species loss (Scherer *et al.*, 2023), but they have not been imported into HPC. The pattern across all three minor categories is the same. The boundary of the literature does not trace the boundary of HPC's environmental footprint. It traces the boundary of what the community has so far built the instruments to measure.

2.7 From Efficiency to Sufficiency

Concentrated on carbon measured during operation, the corpus treats efficiency as the sustainability problem, and that framing has a known limit. Gains in efficiency are repeatedly absorbed by growth in demand: studies of digital systems document rebound effects large enough to cancel the savings (Peng *et al.*, 2023), and analyses of Top500 machines show total electricity consumption still rising even as performance per watt improves (Benhari, Trystram, *et al.*, 2024; Benhari, Denneulin, *et al.*, 2025). Efficiency also leaves whole dimensions untouched. Embodied and other Scope 3 emissions are estimated at roughly 38 to 69 percent of a data center's carbon footprint (Schneider Electric, 2023), yet they sit outside the performance-per-watt metrics that dominate HPC evaluation, together with the water and material burdens the corridor never reaches.

Sufficiency concerns the scientific value of a computation relative to its environmental cost, not only the efficiency with which it runs, and requires metrics such as emissions per validated result rather than per unit of work (Portegies Zwart, 2020; Lannelongue *et al.*, 2021). Groundwork already exists, from the Software Carbon Intensity specification and its functional units (Green Software Foundation, 2024) to frugality frameworks emerging in adjacent sectors (Association Française de Normalisation, 2024).

The gap exposed by the classification can be viewed from two angles. The first concerns the operational side of HPC: the literature provides no mechanism for acting on environmental costs in scheduling, allocation, and replacement decisions. The second concerns HPC systems as a whole: the field evaluates their greenness through efficiency metrics where sufficiency is also needed. These angles define the two research directions pursued in the rest of this document.

Chapter 3

Preliminary Results

This chapter begins with the operational direction through two studies of job timing inside the batch scheduler. The first tests a rule that suppresses backfill starts on a relative grid signal. The second prices candidate start times in absolute impact units and uses an offline analysis to bound what job shifting can achieve. Together, the studies show why refusing starts is insufficient and how much remains available to a scheduler that chooses them.

3.1 Suppressing Backfill Starts on a Relative Signal

The batch scheduler is a practical place to act on operational impacts because it already decides when and where every job runs, but an environmental gain from scheduling is only worth having if the performance penalty stays acceptable. EASY backfilling is a well-established baseline for weighing that trade-off. It keeps a capacity reservation for the job at the head of the queue and raises utilization by filling the resulting gaps with later jobs that do not delay that reservation (Skovira *et al.*, 1996; Mu'alem and D. Feitelson, 2001). The underlying question is whether a simple scheduling heuristic can lower an HPC center's carbon or water footprint without hurting performance. Greenfilling is a first attempt at answering it, and the smallest intervention we could make against the EASY baseline: it leaves the reservation untouched and modifies exactly one decision, permitting a backfill start only when the targeted grid intensity, carbon or water, observed at the scheduling instant does not exceed its exponential moving average. Building and evaluating it doubled as a first exercise of the simulation infrastructure this branch needs.

The rule is event-driven: Greenfilling is implemented as a Batsched variant that subclasses the EASY implementation it is compared against, and the scheduler runs at every job arrival and completion. At the k-th such event, at time t_k, it reads the targeted intensity $I(t_k)$ from the 15-minute trace and updates a smoothed reference

$$S_k = \alpha\, I(t_k) + (1-\alpha)\, S_{k-1}, \qquad S_0 = I(t_0), \tag{3.1}$$

with smoothing factor $\alpha = 0.5$ throughout the campaign. The scheduler permits backfill starts at t_k if and only if $I(t_k) \leq S_k$, and it schedules the reservation for the job at the head of the queue regardless of this condition. Because Equation (3.1) advances once per scheduling event rather than once per trace step, the effective memory of S_k tracks the density of arrivals and completions instead of wall-clock time. When many events fall inside one 15-minute step the reference converges toward the current sample, so how long backfill stays suppressed after an upward step of the signal depends on how busy the machine is, not only on α.

Evaluating Greenfilling first required simulation support that existing toolchains do not offer, since current simulators do not track time-varying carbon and water footprints and a recent extension in this direction covers only operational carbon emissions (Saraiva *et al.*, 2025). We therefore extended SimGrid (Casanova *et al.*, 2025) and Batsim (Dutot *et al.*, 2017) with a per-host footprint model that accumulates operational carbon from the grid carbon intensity, on-site water through the water usage effectiveness (WUE), off-site water through the power usage effectiveness (PUE) and the grid water intensity, and embodied carbon and water as separate values. All four operational parameters can vary independently over time, Batsim replays timestamped updates to them during scheduling experiments, and each job's operational and embodied footprints are exported separately.

The grid signals derive from ENTSO-E actual net generation by production type (Hirth *et al.*, 2018), resampled to a 15-minute grid and converted into generation-weighted carbon and water intensities using lifecycle carbon factors from UNECE where available and IPCC otherwise (United Nations, 2022; Intergovernmental Panel On Climate Change, 2015) and operational water consumption factors (Macknick *et al.*, 2012). Germany, France, and Poland provide three contrasting signals from the same source and observation period, dominated by hard coal and lignite in Poland, by nuclear in France, and by a more mixed and changing supply in Germany. The contrast is not one-dimensional: France has the lowest mean carbon intensity of the three but the highest mean water intensity, 36.4 gCO_2eq/kWh and 3.320 L/kWh against Poland's 569.4 gCO_2eq/kWh and 1.587 L/kWh. A consistency check against Ember's independent production-based national intensities for 2018 to 2024 yields Pearson correlations between annual means of 0.987 for Germany, 0.994 for France, and 0.999 for Poland, with modeled levels 16.9%, 34.0%, and 19.7% lower, a gap that the choice of emission factors largely explains. Since Greenfilling reacts to changes over time rather than to absolute levels, the preserved ordering and annual trend matter more here than the level offsets.

The evaluation campaign pairs these signals with two LANL machines of different scales, Mustang at 1,600 nodes and Trinity at 9,408 nodes, each represented by one slack and one stress workload extract of four weeks selected on utilization and queue conditions. The platform model is deliberately simple: jobs are rigid and replayed at their recorded size, nodes draw 10 W idle and a constant 320 W or 270 W while busy on Mustang and Trinity, and the facility defaults to a PUE of 1, a WUE of 0, and no embodied values, so the campaign reports operational grid carbon and off-site water alone. Constant active power also bounds what scheduling can reach, since reordering the same jobs cannot change active energy and the only remaining levers are idle periods and the operational intensities a job encounters while it runs. Each country contributes 12 environmental windows of 28 days, three drawn per season under a fixed seed, for 36 windows in total. Figure 3.1 shows the final experimental testbed architecture, while Table 3.1 summarizes the parameters of the campaign.

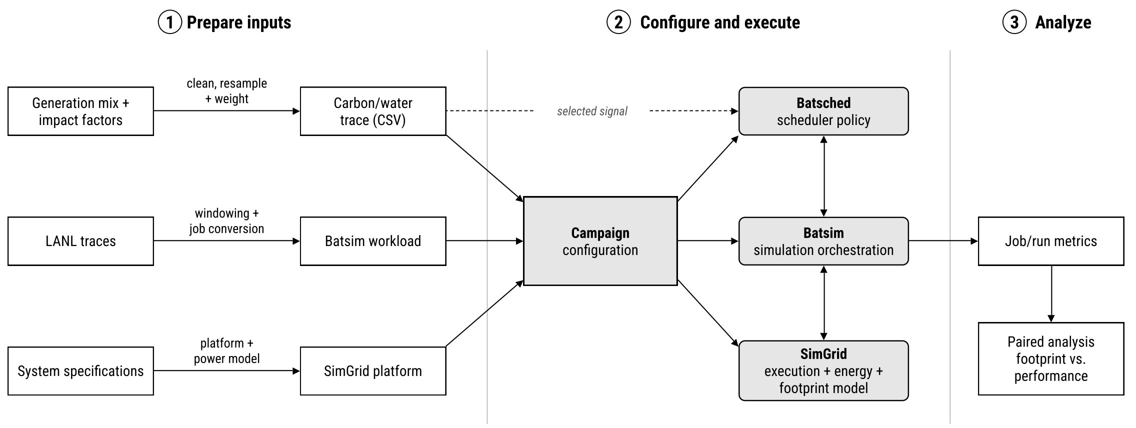

Figure 3.1: *Simulation testbed for the Greenfilling campaign. Preparation converts raw sources into the three artifacts a run consumes, the campaign configuration binds them into a single simulation, and Batsched, Batsim, and SimGrid exchange decisions and events while the footprint model accumulates impacts. The dashed path marks the environmental trace reaching the scheduler directly.*

Varied factors	
Grid signal country	Germany, France, Poland
Environmental window	12 per country, 3 per season, 28 days
Machine	Mustang (1,600 nodes), Trinity (9,408 nodes)
Workload regime	slack, stress
Scheduler	EASY, carbon Greenfilling, water Greenfilling
Fixed settings	
EMA smoothing factor	$\alpha = 0.5$
Grid signal resolution	15 minutes
Node power	10 W idle, 320/270 W busy (Mustang/Trinity)
Facility model	PUE = 1, WUE = 0, no embodied impacts
Total	432 runs, 288 paired comparisons

Table 3.1: *Parameters of the Greenfilling evaluation campaign.*

Crossing windows, machines, workload regimes, and three schedulers, EASY plus a carbon-targeting and a water-targeting Greenfilling, gives 432 simulations, and pairing each Greenfilling run with the EASY run of the same window, machine, and regime gives 288 paired comparisons. Greenfilling is not competitive in any evaluated scenario, that is it does not reduces the targeted operational footprint relative to EASY without increasing mean bounded slowdown. Figure 3.2 shows the paired changes relative to EASY, and Table 3.2 breaks the medians down by machine, regime, and objective. No scenario median footprint change reaches 0.1% in magnitude, for carbon or for water, while mean bounded slowdown rises in every one of the 288 pairs, between +24.5% and +757.2%, with scenario medians between +161.5% and +323.6%. The 95th-percentile waiting time of replayed jobs also rises in all 288 pairs. Even the most favorable case, Mustang under the slack regime, where carbon falls in 30 of 36 windows under carbon Greenfilling and water falls in 29 of 36 under water Greenfilling, buys those reductions at a median slowdown increase of +323.6%. Under stress the Mustang medians turn slightly positive and Trinity stays near zero in both regimes. Zero of the 288 pairs reduce the targeted footprint without increasing mean bounded slowdown.

Machine	Regime	Objective	Footprint	Energy	Slowdown
Mustang	slack	carbon	−0.068%	0.000%	+323.6%
		water	−0.031%	0.000%	+308.5%
	stress	carbon	+0.008%	+0.043%	+175.0%
		water	+0.014%	+0.044%	+161.5%
Trinity	slack	carbon	−0.005%	0.000%	+210.1%
		water	−0.007%	0.000%	+211.5%
	stress	carbon	+0.001%	0.000%	+226.6%
		water	−0.000%	0.000%	+213.3%

Table 3.2: *Median paired changes relative to EASY by machine, workload regime, and Greenfilling objective. The footprint column reports the signal each objective targets. On Mustang under slack the consumed platform energy is identical to EASY in every window, and on Trinity it is unchanged to numerical precision at the median.*

The energy column of Table 3.2 separates the two ways a footprint can change under this platform model, consuming a different amount of energy or meeting a different intensity while consuming it. On Mustang under slack and on Trinity in both regimes the consumed energy does not move, so the footprint change there is purely an exposure change, and that exposure change stays below 0.1%. Mustang under stress is the one exception. Median platform energy rises by 0.043% under carbon Greenfilling and 0.044% under water Greenfilling, while the effective targeted intensity, the targeted operational footprint divided by consumed energy, falls by 0.037% and 0.032%. The rule does buy a small exposure improvement there, but the extra energy from stretching the schedule cancels it, which is how the stress medians on Mustang end up slightly positive. The campaign accounts energy at the whole-platform level, so this decomposition shows the mechanism but cannot attribute the extra energy to individual jobs.

The design itself points at why deferral did not translate into lower footprints. Greenfilling decides whether a backfill may start at the current instant, but it never selects a later start based on predicted intensity. Under constant active power, postponing a job changes its footprint only if its execution actually moves to lower-intensity intervals, and the campaign does not determine whether that condition failed because of the decision rule, the temporal structure of the signals, the workloads, or their interaction. The campaign does probe the signal hypothesis once. Each window's swing, the mean daily peak-to-trough range of the targeted signal divided by the window mean, correlates only weakly with the achieved saving, at a pooled Pearson $r = 0.079$ for carbon and $r = 0.247$ for water over 144 pairs per objective, with per-scenario coefficients between −0.09 and 0.59 and no consistent sign. Every window is reused by all four machine and regime scenarios, so the pooled points are not independent, but windows with more intra-day variation did not systematically yield larger savings. The result is therefore narrow: under the evaluated

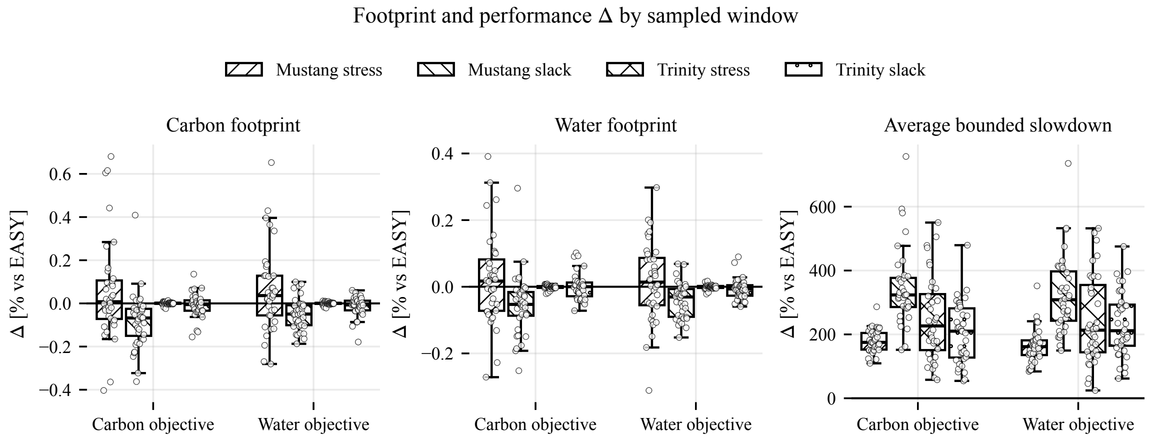

Figure 3.2: *Footprint and performance change relative to EASY across the 288 paired comparisons. Carbon and water footprint changes stay small, with medians below 0.1% in magnitude, while mean bounded slowdown worsens in every pair.*

signals, workloads, and platform assumptions, suppressing EASY's backfill starts on a moving-average comparison does not reduce the targeted footprint enough to offset its increase in mean bounded slowdown.

Two lessons from the campaign now shape the mechanism that replaces it. The first is the asymmetry between refusing a start and choosing one. A suppressed backfill start buys a certain queueing delay for an environmental return that depends on the intensities the deferred job eventually meets, and the campaign shows how one-sided that bet is. The second is that a relative threshold and a relative metric compound each other. The rule fires on the same relative deviation in France as in Poland even though their mean carbon intensities differ by more than an order of magnitude, a 0.1% saving reads identically in both, and the quantity that would distinguish them, the absolute footprint avoided per unit of scheduling penalty, was not measured. Together, these lessons set a lower bound: an intervention that only refuses starts on a relative signal is too little. A viable heuristic has to price both sides of each decision, the footprint of starting a job now against the footprint of waiting, and it has to weigh absolute rather than relative stakes.

The campaign also produced reusable infrastructure. We release the intensity traces, workload extracts, campaign configuration, and analysis as a reproducible artifact,[1] and we submitted this analysis to the EESP 2026 workshop.

3.2 Pricing Each Start in Absolute Terms

The Greenfilling lessons point to job shifting as the next mechanism to explore. Rather than suppressing opportunistic starts on a relative comparison, the scheduler could choose each job's start time by pricing every candidate start within a bounded delay, in the absolute units of the targeted impact.

The model prices a single job: let j be a job with execution time p_j, q_j allocated nodes,

[1] https://github.com/Lucas-Doctorate-Project/experimental-artifacts/tree/paper

and submission time r_j, and let I be the time-varying intensity of the targeted signal. Starting j at a time $t \geq r_j$ costs

$$C_j(t) = P_{\text{idle}}\, q_j \int_{r_j}^{t} I(\tau)\,\mathrm{d}\tau + P_{\text{comp}}\, q_j \int_{t}^{t+p_j} I(\tau)\,\mathrm{d}\tau, \tag{3.2}$$

where P_{idle} and P_{comp} are the platform's idle and busy node powers. The first term charges for the wait: while j holds back, its q_j nodes sit idle and accumulate impact at the intensities of the interval $[r_j, t)$. The second term charges for the run, integrating the intensity over the window the execution would actually occupy rather than sampling it at the start. The heuristic selects the start t^* that minimizes C_j over $r_j \leq t \leq r_j + d_m$, where the maximum displacement d_m bounds how long any job can be held. At $t = r_j$ the waiting term vanishes and Equation (3.2) reduces to the cost of starting immediately, so not shifting is itself a candidate and the heuristic only moves a job when the trace ahead pays for the wait. Carbon and water each define their own cost function through their own intensity series, and thus their own t^*. The cost of deferral appears as idle exposure, the return appears as the difference in integrated intensity over the execution window, and both sides are absolute quantities rather than deviations from a moving average.

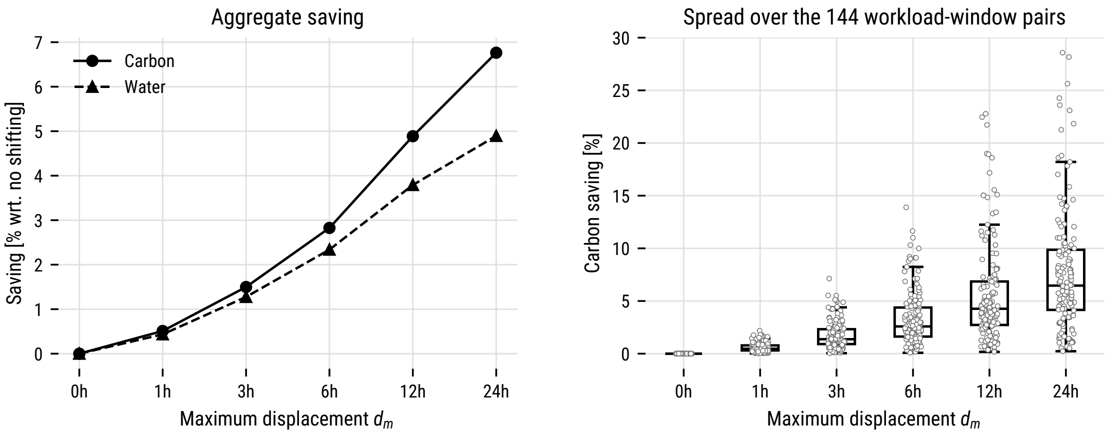

Figure 3.3: *Ceiling of the offline job-shifting analysis as a function of the maximum displacement d_m. The left panel shows the aggregate saving over all workloads and windows for each signal. The right panel shows the spread of the carbon saving across the 144 workload and window pairs.*

Before the heuristic reaches the scheduler, the same cost function supports an offline analysis of the workloads and traces. The grid signals are step functions on a 15-minute grid, so the integrals in Equation (3.2) collapse into sums. We use this to compute a ceiling: for each job, excluding the warm-up jobs that reconstruct the machine state at the start of each extract, we record the saving it would collect if it could shift freely within d_m, and we accumulate these per-job gains over the same 144 workload and window combinations as the campaign, with the same platform power values, for d_m between zero and 24 hours. The ceiling is optimistic by construction, since summing independent per-job gains ignores node capacity and queue contention and lets every job claim the same low-intensity valleys at once. Its value is diagnostic: if even this bound were small, no shifting heuristic on these signals would be worth implementing.

The bound is not small, as Figure 3.3 shows. The aggregate ceiling grows with the

Grid	Signal	Mean	Median	Std	Min	Max
Germany	carbon	0.50	0.45	0.22	0.29	0.93
	water	0.47	0.39	0.21	0.27	0.84
France	carbon	0.32	0.30	0.08	0.18	0.51
	water	0.29	0.27	0.07	0.19	0.44
Poland	carbon	0.24	0.20	0.15	0.09	0.61
	water	0.23	0.19	0.15	0.10	0.60

Table 3.3: *Swing of the carbon and water signals by grid, over the 12 windows per country used in the Greenfilling campaign.*

allowed displacement and reaches 6.76% for carbon and 4.89% for water at a d_m of 24 hours, almost two orders of magnitude above the sub-0.1% medians of the Greenfilling campaign. Water pays off less than carbon over the long run, likely due to its lower day-to-day variability. Averaged over the 12 windows per country used in the Greenfilling campaign, mean swing is 0.50 for carbon against 0.47 for water in Germany, 0.32 against 0.29 in France, and 0.24 against 0.23 in Poland (Table 3.3).

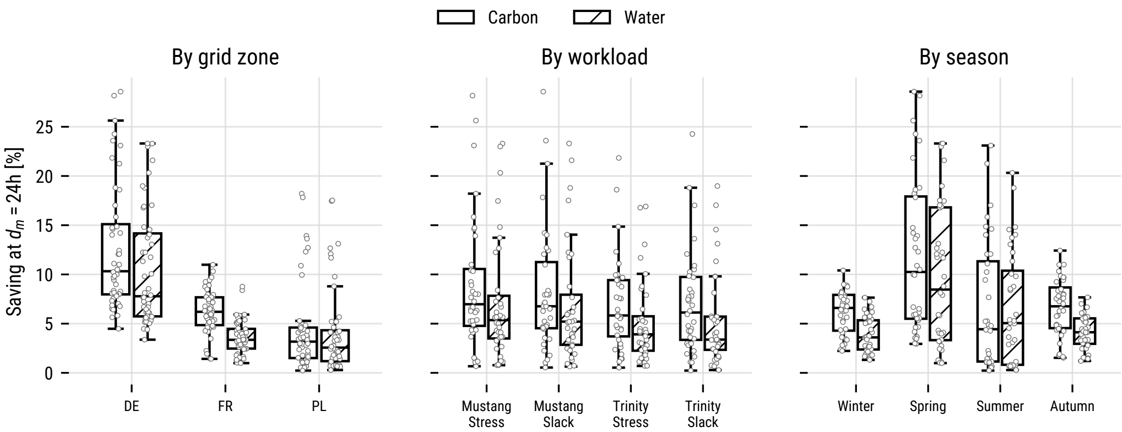

Figure 3.4: *Ceiling at a d_m of 24 hours across the 144 workload and window pairs, split by grid zone, workload scenario, and season.*

Within that bound, the gains concentrate by grid and by season (Figure 3.4). The German signal offers the most, with an aggregate ceiling of 11.69% at 24 hours, against far less in France and Poland, and the seasons follow the same mechanism: spring reaches 12.32% against winter's 4.89%. The gap tracks the same swing metric: over the three windows drawn from each of the three grids, mean carbon swing is 0.50 in spring against 0.28 in winter, nearly double, so a job displaced in spring meets a wider daily range between low and high intensity than one displaced in winter, and that wider range is what the heuristic exploits. The four workload scenarios behave similarly, which points at the signal rather than the workload as the driver, though with only three windows per grid and season the seasonal split is still a tendency rather than a settled result.

Only a fraction of jobs actually move (Figure 3.5): at a d_m of 24 hours, 22.2% of jobs

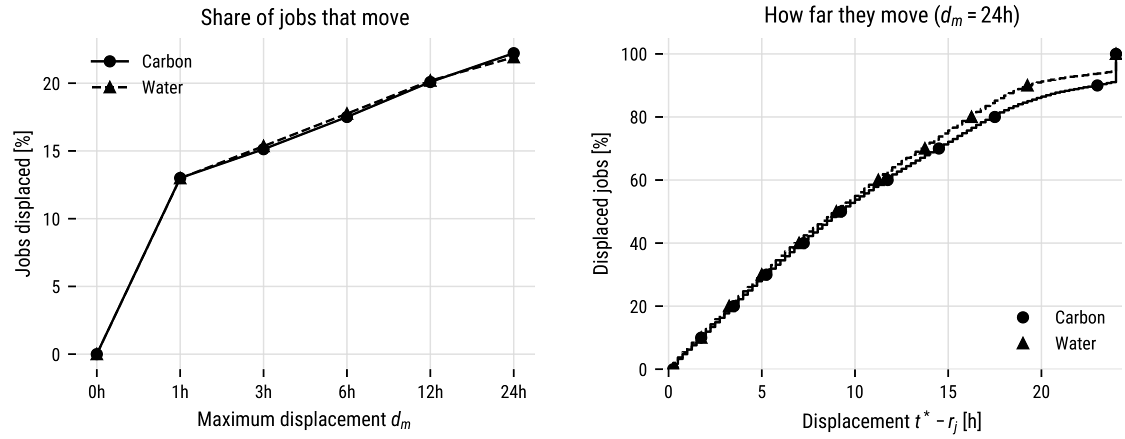

Figure 3.5: *Share of jobs the offline analysis displaces as a function of d_m (left) and the distribution of the displacement $t^* - r_j$ among displaced jobs at a d_m of 24 hours (right).*

are displaced at all, with a median displacement of 9.25 hours among those that move, while the other 78% already start at the best point they can reach. The aggregate ceiling is therefore carried by roughly a fifth of the jobs, which sharpens the capacity caveat, since the bound assumes all of them fit into the same valleys at the same time.

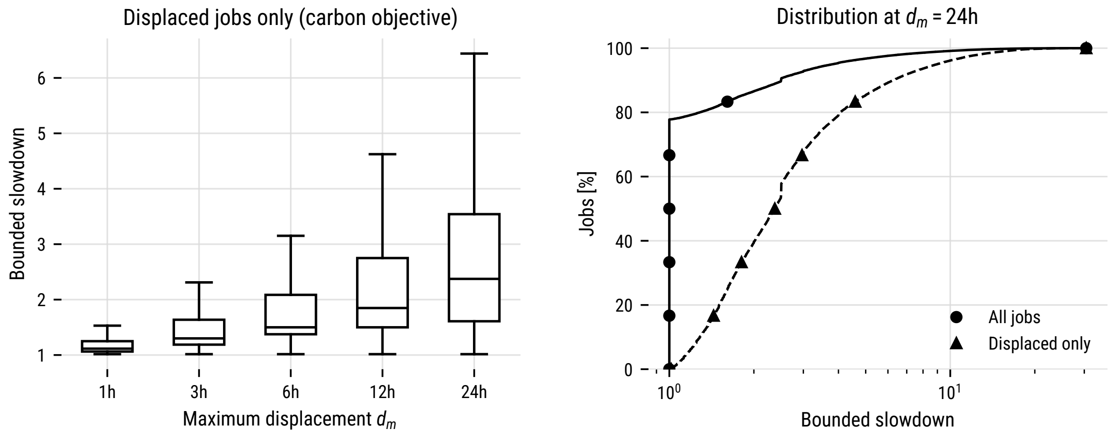

Figure 3.6: *Bounded slowdown implied by the displacements under the carbon objective. The left panel shows displaced jobs only, by maximum displacement. The right panel shows the cumulative distribution at a d_m of 24 hours for all jobs and for displaced jobs only.*

Shifting a job deterministically adds $t^* - r_j$ to its waiting time, so its bounded slowdown, $\max\big(1, (t^* - r_j + p_j)/\max(p_j, \tau)\big)$ with a bounding threshold of $\tau = 10$ s, rises with every displacement. Under the carbon objective at a d_m of 24 hours the mean bounded slowdown is 1.49 across all jobs and 3.22 among displaced jobs, with a 99th percentile of 15.4 (Figure 3.6). Saving and delay also move together across scenarios: per workload and window pair, the carbon saving correlates with the mean bounded slowdown at $r = 0.874$ (Figure 3.7). The windows that save the most are the windows where jobs wait the longest, so the trade-off is still there, but unlike under Greenfilling each unit of delay now buys a measured environmental return.

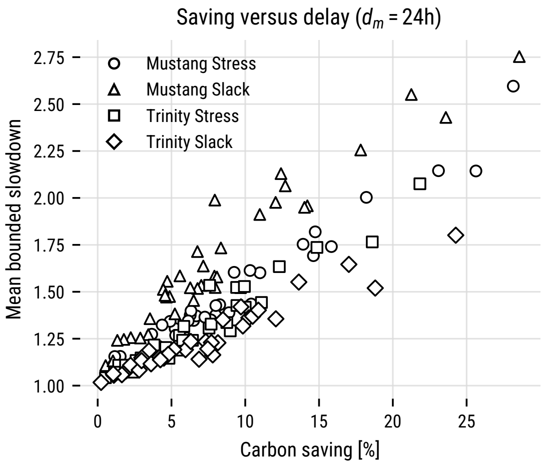

Figure 3.7: *Carbon saving against mean bounded slowdown at a d_m of 24 hours, one point per workload and window pair.*

Chapter 4

Research Plan

This chapter presents the research plan through the thesis defense. Section 4.1 presents what to expect from the job shifting experiments and how we plan to develop mechanisms involving the scheduler, users, platform controls, and hardware replacement. Section 4.2 presents the sufficiency-aware evaluation of public machine data.

4.1 Accounting for Environmental Cost in HPC Operations

Greenfilling showed what is not enough. Refusing a start when the current intensity sits above its moving average buys a certain delay for a return that depends on intensities the deferred job may never meet. Chapter 3 reports the outcome: no scenario median moved the targeted footprint by 0.1%, and mean bounded slowdown rose in all 288 pairs. A policy therefore has to price both sides of the decision it makes, the impact of acting now against the impact of waiting.

An HPC center's cost can derive from four entities, shown in Figure 4.1. Users typically submit work and are charged in core-hours. The scheduler chooses when and where each job runs based on queue state and node availability. The platform run its nodes at fixed power and frequency settings. Hardware is often replaced on a cycle determined by performance and procurement considerations. Environmental cost rarely enters any of these decisions.

The rest of this section brings environmental cost into each of these decisions, one step at a time. The first step stays with the scheduler and closes the job shifting arc opened in Chapter 3. The second brings in the platform, turning its power settings into something the scheduler can price alongside start time. The third moves past day-to-day operation into hardware aging and replacement. The fourth reaches the users, since they decide what work arrives and how much delay it tolerates.

Job shifting is already known in the literature. Wiesner et al. (Wiesner *et al.*, 2021) report savings from temporal shifting in the cloud, and Sukprasert et al. (Sukprasert *et al.*, 2024) report how quickly those savings shrink once deadlines and workload structure

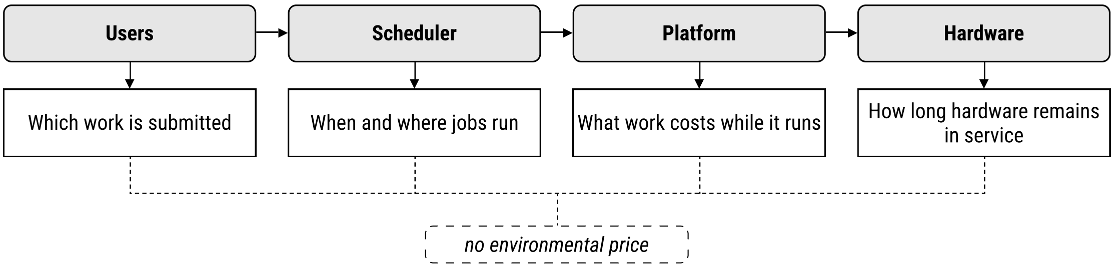

Figure 4.1: *Entities that dictate the operational cost of HPC. Users decide what work is submitted, the scheduler decides when and where it runs, the platform sets what running it costs, and hardware replacement decides how long the machine stays in service. These decisions rarely prices the environmental cost.*

are respected. Equation (3.2) prices a single job in isolation, and the offline analysis of Chapter 3 used it to bound what shifting can reach, 6.76% of operational carbon and 4.89% of water at a maximum displacement of 24 hours. That bound is unrestricted: it lets every one of the 22.2% of displaced jobs claim the same low-intensity valleys at once, and it assumes the whole intensity trace is known at the moment of the decision. Implementing the cost function as a Batsched scheduler removes the first assumption, since node capacity then decides which jobs actually reach the valleys they price. We will evaluate it against EASY and both Greenfilling variants on the campaign design of Table 3.1, sweeping d_m, and report what fraction of the offline bound survives contention and what that fraction costs in bounded slowdown.

The question becomes how much of the realized saving survives when the cost function integrates a forecast instead of the recorded signal. The ENTSO-E platform the campaign draws its signals from also publishes day-ahead generation forecasts (Hirth *et al.*, 2018), so the data is ready to use. This removes the second assumption and closes the questions left open in Chapter 3.

The second step starts from a limitation of the platform model: it holds active power constant, so reordering the same jobs cannot change active energy. The only levers left are idle periods and the intensities a job meets while it runs. However, there are other levers not explored: node power states, as well as per-node DVFS and RAPL power caps (David *et al.*, 2010). EcoFreq (Kozlov and Stamatakis, 2024) scales power against carbon and price signals, and it gives us a reference point.

Equation (3.2) currently prices only a job's start time. Adding power state as a second decision variable lets the scheduler price both together and trade execution time against power as well as delay. That poses a question the earlier steps cannot ask: does slowing a job down beat making it wait? Running at a lower frequency stretches a job across more of the trace and lowers its draw throughout, while waiting preserves the draw and moves it. Neither is obviously better, and which one wins should depend on the shape of the signal and on how a job's runtime responds to frequency.

Cooling also provides a second control surface beyond the compute nodes. The footprint model already supports time-varying PUE and WUE, but the Greenfilling campaign left these values fixed, leaving this capability unused. Making cooling a decision variable would require a thermal model that the testbed currently lacks. We will prioritize compute controls first and investigate cooling controls only if the results show that the trade-off between execution time and power consumption justifies the additional model complexity.

The third step leaves the operational stage for replacement timing, which decides how much manufacturing impact a center commits to and how long it has to amortize that impact. Li et al. (Li *et al.*, 2023) and Uwizeyimana and Enright Jerger (Uwizeyimana and Jerger, 2025) frame the decision as a crossover: a newer machine lowers operational impact per unit of work and adds embodied impact at once, so replacing early spends carbon to save carbon, and the balance depends on how the machine is used.

The crossover weighs the embodied impact of a new machine against the operational impact of keeping the current one running. Prior work already estimates the latter for a fixed workload. We want to adapt this idea to footprint-aware scheduling, delaying the crossover and extending the machine's service life when the trade-off justifies it. This turns what is usually an operational decision into a lifecycle-aware one.

The testbed already models embodied carbon and water, so a replacement policy can reason directly about these impacts, amortized over the useful work the machine delivers. Since replacement decisions play out over several years, we will also need a long-horizon simulation setup that captures hardware aging.

The fourth step makes the user an active participant. The user declares how much delay a job tolerates, whether it accepts a reduced power state, and whether the job is submitted at all. Kamatar et al. (Kamatar *et al.*, 2025) surveyed 300 HPC users and found that fewer than 30% were aware of the energy their jobs consume. Work on parallel system traces established that submission behavior responds to how the system performs (Shmueli and D. G. Feitelson, 2007). Other works attribute energy and emissions to individual jobs and users, from resource manager monitoring stacks (Paipuri, 2025) to user-centric footprints computed from job records (Wassermann *et al.*, 2024; Niewenhuis *et al.*, 2024). The Fugaku Points program goes further and rewards users for applying power-control knobs (Solórzano *et al.*, 2024).

At this stage the platform can also respond to the user with a price, a reward, or both, and we want to understand how user decisions change when such a policy is introduced. Price and reward can be independent knobs. A reward can carry no budget of its own, or it can be funded by what the price collects, so that users who defer are paid by users who do not.

We will compare mechanisms on the footprint they remove and on the cost they impose on users in delay and refused work. Since no trace records how users respond to a price, we will model synthetic users and their behavior, which lets us evaluate different scenarios and identify which factors dominate in each case.

The four steps group complex interactions that exist in real systems. The scheduler places work, the platform sets what that work costs, and users decide what work arrives and how much delay it tolerates. The goal with this branch is to understand and provide insights on how these entitites interact.

4.2 Sufficiency-Aware Evaluation of HPC Systems

The mapping study in Chapter 2 showed that evaluation in the field is focused on operational efficiency. The Green500 ranks supercomputers by performance per watt, measured on a single benchmark (Feng and Cameron, 2007). That metric gave the field a shared target beyond raw speed, and the list has run for fifteen years on it (Adhinarayanan and Feng, 2025). A ratio of one impact to one measure of useful work still leaves most of a system's footprint outside the frame. Karanikolaou and Bekakos (Karanikolaou and Bekakos, 2022) find systems of entirely different size and capability sitting on consecutive rows of the Green500. Rao and Chien (Rao and Chien, 2025) put numbers on the same effect in the Top500. Leonardo emits 4.3 times the operational carbon of LUMI, the system ranked directly above it, and Frontier carries 2.6 times the embodied carbon of El Capitan. Absolute power draw, hardware composition, the carbon intensity of the local grid, and the water spent generating that electricity all separate those machines, and a performance-per-watt ratio reports none of it.

The aim is to use the ranking as a reference point and show what a sufficiency-aware lens reveals once that ranking is held against the rest of a system's footprint. Efficiency measures how much useful work a system returns per unit of impact, a ratio that improves as the numerator grows. Sufficiency asks instead whether the total burden a system imposes, and the growth in demand behind it, can be brought down at all (Santarius *et al.*, 2022). A relative metric can improve while the absolute quantity it divides keeps climbing, which is how a fleet grows greener per operation and heavier in total at the same time (York *et al.*, 2022).

The Top500 already sits in that position: performance per unit of carbon keeps improving while the fleet's operational carbon grows by 10.3% a year, on track to nearly double between 2024 and 2030 (Rao and Chien, 2025). The same arithmetic holds outside the machine room. Yang and Wang (Yang and Wang, 2023) compare a Chinese national supercomputing center against a synthetic control city and find that it worsened regional carbon emissions, because growth in energy demand outweighed what the center returned through a cleaner energy structure or greener technology. We propose a diagnostic lens that surfaces what efficiency-only evaluation hides, a step towards deciding how much computation is justified.

We plan to approach this gradually, in three steps that move further from efficiency toward sufficiency. For each one we evaluate what it reveals and what data it demands, and what it trades away. Figure 4.2 lays the three steps against the boundary of what public data supports.

The first step keeps the shape of a ranking but widens the impacts it accounts for. Carbon modelling already reaches both the operational and the manufacturing stage (Li *et al.*, 2023). Water can be estimated from the electricity a system draws and from the scarcity of the region supplying it (Jiang *et al.*, 2025). Beyond performance per watt alone, systems will be ranked according to the impact dimensions presented by the mapping.

The second step will introduce sufficiency-oriented indicators, including, but not limited to, a system's total lifecycle burden, how heavily it is actually used, and how often it is replaced. These indicators are where the efficiency account and the sufficiency

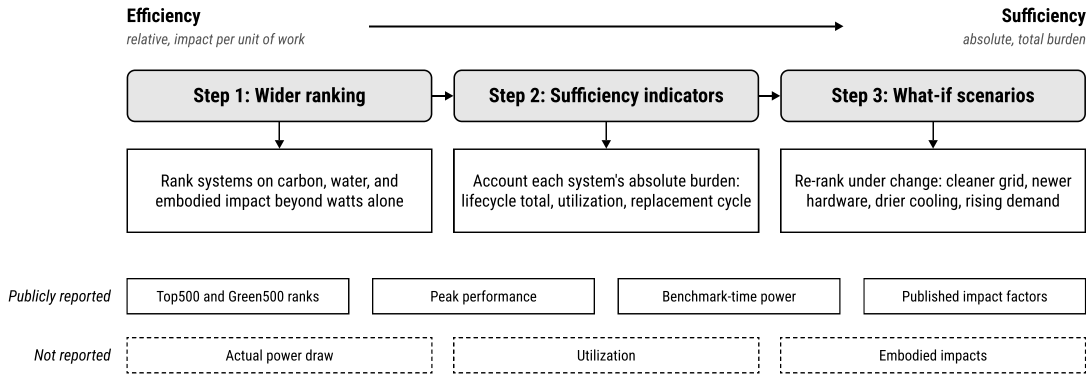

Figure 4.2: *The three evaluation steps move from a wider efficiency ranking toward an account of absolute burden. The public-data boundary cuts across the whole plan rather than any single step: ranked entries, peak performance, benchmark-time power, and published impact factors are on the record, while actual power draw, utilization, and embodied impacts are not.*

account diverge.

The third step turns the missing data into the object of study. Rather than rank systems only as they stand, we will test what the ranking becomes under a change: the same machine on a cleaner grid, with a newer generation of hardware, under a less water-intensive cooling design, or against a rising demand trajectory. Each scenario supplies an assumed figure where the real one is unavailable, then shows how the re-ranking responds.

The challenge with these steps is data. Rao and Chien (Rao and Chien, 2025) found a utilization figure for three of the 500 ranked systems, and essentially none of them report actual power draw. They also show that, once public sources beyond the list are added, operational carbon can be estimated for 98% of the ranked systems and embodied carbon for 80.8%. We expect to resort to estimation and modelling. Beyond the numbers themselves, we intend to provide guidelines on how to make this information more accessible.

Taken together, the three steps move from a wider efficiency ranking toward an account of absolute burden. They rest on the same foundation as the rest of this thesis, the map of Chapter 2 and the premise that HPC's impacts run past energy efficiency. The Green500 gave HPC a way to reward efficient machines. We aim to give it a way to see what efficiency alone leaves out, and to make how much computing is enough part of how systems are judged.

Chapter 5

Schedule and Expected Contributions

This chapter turns the plan of Chapter 4 into a timeline and a list of deliverables. Section 5.1 allocates the steps of both branches over the two years between this qualification and the thesis defense. Section 5.2 lists what the thesis has produced so far and what it commits to produce.

5.1 Schedule

Figure 5.1 lays the steps of Chapter 4 over eight quarters, from September 2026 to August 2028. The branches run in parallel throughout. The job shifting arc comes first, since Chapter 3 left it one implementation away from an answer. The first step of the sufficiency branch follows one quarter later, once the shifting work is underway, and it stays near-term because it needs only public Top500 entries and published impact factors. The later steps of each branch follow the order Chapter 4 gives them. Replacement timing overlaps the end of the power state work because both build on the same simulation infrastructure, and the user mechanisms close the operations branch in the second year, starting as the what-if scenarios wind down.

The two milestones in the middle of the timeline are paper submissions, one per branch. The first collects the operations results available by the end of the first year, the online shifting evaluation and the power state pricing. The second collects the wider ranking and the sufficiency indicators. We plan to submit both to high impact venues in the field, such as Supercomputing, ISC, and SBAC-PAD. The final six months are reserved for writing the thesis, which by then assembles chapters that already exist in some form as papers, analyses, and released artifacts.

5.2 Expected Contributions

Table 5.1 lists the contributions of this thesis, separating what is already delivered from what the schedule above commits to. The mapping study is published in Computer Science Review (Rosa *et al.*, 2026). The footprint model and the Greenfilling analysis exist and support everything the operations branch plans next, and the analysis is under review

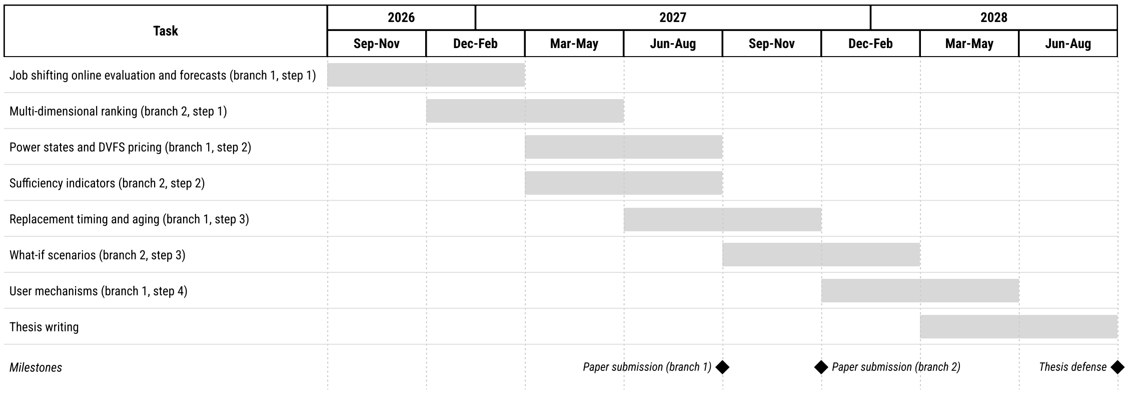

Figure 5.1: *Schedule from the qualification to the thesis defense, in quarters. Each task carries the branch and step it implements from Chapter 4. Diamonds mark the two planned paper submissions and the defense.*

Contribution	Branch	Status
Systematic mapping study (Rosa *et al.*, 2026), Elsevier CSR	both	published
Footprint model for SimGrid and Batsim	1	done
Job shifting analysis, EESP 2026	1	submitted
Reproducible experimental artifact	both	ongoing
Co-supervision of two undergraduate students	1	ongoing
Operations branch paper	1	planned
Sufficiency evaluation paper	2	planned
Upstreaming the footprint model into SimGrid and Batsim	1	planned

Table 5.1: *Contributions of this thesis. Branch 1 is the operations branch of Section 4.1 and branch 2 the sufficiency evaluation of Section 4.2.*

at the 3rd International Workshop on Energy Efficiency with Sustainable Performance (IEEE/ACM SC EESP Workshop 2026). The experimental artifact is ongoing work that accumulates the traces, configurations, and analyses of each campaign as the research advances. Two undergraduate students work under our co-supervision on the preliminary results line, extending the shifting analysis. Once the footprint model and the simulation infrastructure around it are stable, we plan to upstream them into SimGrid and Batsim rather than maintain a fork, so the infrastructure outlives the thesis.

These deliverables also carry contributions beyond their immediate results. For the field, the mapping study and the sufficiency evaluation widen how systems are judged beyond performance per watt, and the data boundary drawn in Section 4.2 tells operators and vendors exactly which figures are missing and what publishing them would support. For practitioners, the mechanisms of the operations branch are candidate policies for schedulers and resource managers already in production, and their evaluation reports what every operator asks first, how much footprint a policy removes and how much slowdown it costs. For researchers, the footprint model, the traces, and the campaign tooling lower the cost of entering this line of work, and upstreaming them into SimGrid and Batsim keeps that infrastructure maintained after the thesis ends.

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